

# PA2 Report - Requirement 2: User Analysis

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**Group:** 06

**Course:** CSC13112 - UI/UX Design

**Assignment:** Project Assignment 2

**Product Scope:** Freestyle Chess mobile website on smartphone browser

**Document Purpose:** User problem identification, idea generation, affinity diagramming, and problem prioritization

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## 1. Overview & Brainstorming Setup

### 1.1 Objectives

The goal of this phase is to analyze the empirical evidence collected from Requirement 1 (72 survey responses and 13 direct interview sessions), brainstorm to identify core user problems, generate potential solution ideas, organize them using an **Affinity Diagram**, and prioritize the most severe problems to solve in the subsequent design proposal.

### 1.2 Brainstorming Session Record

| Field              | Details                                                                                                                                               |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| Date & Time        | July 23, 2026                                                                                                                                         |
| Participants       | Group 06 Members (Lê Mai Hoài Bảo, Trương Công Thiên Phú, Lâm Hữu Khánh, Phạm Chí Bảo Ninh, Phùng Ngọc Tuấn)                                          |
| Tool Used          | Digital Sticky Notes via Figma / FigJam                                                                                                               |
| Input Evidence     | 06-PA2-UserResearch.md, Khảo sát trải nghiệm người dùng trên Freestyle Chess mobile web (Responses) – Form Responses 1.csv, Interview clip recordings |
| Deliverable Output | Affinity Diagram & Prioritized Problem Matrix                                                                                                         |

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## 2. Affinity Diagram

### 2.1 Visual Affinity Board (Digital Sticky Notes)

Below is the structured Affinity Board generated during the group brainstorming session, grouping user observations, pain points, and potential solution ideas into logical clusters.



Link to interactive FigJam board: [FigJam Affinity Diagram Canvas](#)

## 2.2 Raw Sticky Notes Inventory

**Notes:** Data in the "Source / Evidence" column supports the "Stick Note Content", exceptions are listed below the table.

| Note ID | Category | Source / Evidence                           | Sticky Note Content                                                                                      |
|---------|----------|---------------------------------------------|----------------------------------------------------------------------------------------------------------|
| N01     | Problem  | Survey Q12, P01/P02/P03/P04/P05/P07/P08     | Mobile Hamburger menu is placed at top-left, making one-handed thumb interaction painful/annoying.       |
| N02     | Problem  | Survey Q13 (43.1%), P02/P03/P04/P05/P07/P08 | Users must stretch fingers, shift phone grip, shift the phone, or use two hands to tap navigation items. |
| N03     | Idea     | Brainstorm                                  | Move primary navigation to a persistent Bottom Navigation Bar.                                           |
| N04     | Idea     | Brainstorm, P05                             | Provide a floating action button (FAB) or bottom-right drawer for one-handed reachability.               |

| Note ID | Category | Source / Evidence                           | Sticky Note Content                                                                                                  |
|---------|----------|---------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| N05     | Problem  | Survey Q09 (avg 3.68/5), P03/P05/P08        | Homepage Hero section is cluttered with banners and lacks onboarding explaining Freestyle Chess format to new users. |
| N06     | Problem  | Survey Q10 (51.4%), P03/P05/P08             | New users feel disoriented and do not understand website purpose upon landing.                                       |
| N07     | Idea     | Brainstorm                                  | Redesign Hero section with a minimalist intro banner, clear onboarding text, and prominent CTA.                      |
| N08     | Idea     | Brainstorm                                  | Add an interactive "What is Freestyle Chess?" quick modal guide for newcomers.                                       |
| N09     | Problem  | Survey Q17 (81.9%), P02/P05/P06/P08         | Rating leaderboard is an infinite scroll list with no sticky search bar, causing repetitive scrolling.               |
| N10     | Problem  | Survey Q19 (61.1%), P02/P04/P06/P07         | Users cannot search news/events by keywords and must browse long chronological lists.                                |
| N11     | Idea     | Brainstorm                                  | Add a Sticky Search Bar fixed at top of long listing pages (Rating, News, Videos).                                   |
| N12     | Idea     | Brainstorm                                  | Provide filter chips (by major event, date, rating range) and auto-suggest search queries.                           |
| N13     | Problem  | Survey Q21 (52.8%), P02/P03/P05/P06/P07/P09 | Video player lacks immediate visual feedback (loading spinner) when tapped, leading to rage clicks.                  |
| N14     | Problem  | Interview P09/P10 clips; P04/P05/P07        | Playing a new video does not pause the previous video, resulting in overlapping audio channels.                      |
| N15     | Idea     | Brainstorm; P04/P05/P07/P09                 | Enforce a Single-Player audio/video model (playing new video automatically stops old video).                         |
| N16     | Idea     | Brainstorm; P02/P03/P09                     | Add skeleton loaders and loading spinners instantly upon tapping video play controls.                                |

#### Exceptions:

- P08 Neutral about N10.

- P08 supports N15 but neutral about N14.
- P06 doesn't think N14 is a problem, and actually like it that way.

### Additional Problem Outside the Survey:

**Notes:** These are problems that arose from direct Interviews rather than from questionnaires, worth considering.

| Note ID | Category | Source / Evidence | Sticky Note Content                                                                                                                                                                              |
|---------|----------|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| N17     | Problem  | P01               | Confirmation dialogs are too small, forcing users to scroll inside the dialog to read the full message before making a decision.                                                                 |
| N18     | Problem  | P01               | Page transitions provide no loading indicator, making users unsure whether the site is loading or frozen and encouraging repeated taps.                                                          |
| N19     | Problem  | P02               | Video & Streams carousel looks swipeable because thumbnails are cut off, but horizontal swiping does not work and users must use Prev/Next buttons instead.                                      |
| N20     | Problem  | P02               | Schedule navigation uses a down-arrow icon suggesting expansion, but the content slides in from the side, causing confusion.                                                                     |
| N21     | Problem  | P02               | Schedule has two different exit controls ("X" and "Back to Main Menu"), making it unclear which one to use.                                                                                      |
| N22     | Problem  | P05               | Rating list rows have overly similar typography/visual structure, making it easy to overlook the desired player while scanning.                                                                  |
| N23     | Problem  | P06/P09           | Rating search requires the player's name in a specific format/order, causing searches to fail when users enter a natural first-name/last-name order.                                             |
| N24     | Problem  | P09               | Schedule does not provide enough information about upcoming matches or tournament brackets, forcing users to look elsewhere for this information.                                                |
| N25     | Problem  | P09               | Video carousel stops responding to swipe gestures once a video is playing, forcing users to use Prev/Next controls instead.                                                                      |
| N26     | Problem  | P10               | Hero video "Stop" button does not actually stop the video OR restart the video from the beginning instead of correctly pausing/stopping it. Button state does not match the video's actual state |
| N27     | Problem  | P10               | Ranking search cannot search by chess title or rating, despite users naturally wanting to find players using those attributes.                                                                   |
| N28     | Problem  | P10               | Schedule banners appear interactive/tappable but do nothing when tapped.                                                                                                                         |

| Note ID | Category | Source / Evidence | Sticky Note Content                                                                                    |
|---------|----------|-------------------|--------------------------------------------------------------------------------------------------------|
| N29     | Problem  | P10               | Schedule menu has a broken scroll boundary that prevents users from fully viewing some of its content. |

## 2.3 Affinity Cluster Descriptions

### Cluster A: Ergonomic Navigation & One-Handed Reachability

- **Core Problem:** The current hamburger menu icon is positioned at the top-left corner. On modern smartphone screens, this forces right-handed users (44.4% one-handed users) to stretch their thumb unnaturally or change their grip, increasing device drop risk.
- **Generated Ideas:**
  - Implement a mobile **Bottom Navigation Bar** with key tabs (Home, News, Schedule, Rating, Videos).
  - Place a floating drawer or menu button at the bottom-right corner.

### Cluster B: Homepage Onboarding & Value Proposition Clarity

- **Core Problem:** 52.8% of surveyed users had never heard of Freestyle Chess. The current homepage Hero section is overloaded with event banners without explaining the unique rules/format of Freestyle Chess, causing cognitive fatigue.
- **Generated Ideas:**
  - Simplify the Hero section hierarchy.
  - Add a prominent onboarding banner with a concise explanation ("Fischer Random / Chess 960 format") and a primary CTA ("Explore Events").

### Cluster C: Search, Filtering & Information Retrieval

- **Core Problem:** 61.1% of users reported difficulty finding specific info when they don't remember exact names. Long rating lists require 20+ seconds of vertical scrolling without a sticky search bar or pagination.
- **Generated Ideas:**
  - Add a **Sticky Search Bar** that remains visible while scrolling.
  - Implement **Quick Filter Chips** (e.g., "Geller Cup", "Grandmaster Rating") and Auto-suggest queries.

### Cluster D: Video Player Interactions & Feedback

- **Core Problem:** Tapping video thumbnails provides no loading indicator, causing users to think the site is frozen. Multiple videos can play simultaneously, causing overlapping audio streams.
- **Generated Ideas:**
  - Enforce **Single-Active-Player logic** (auto-pause active video when another starts).
  - Integrate skeleton loaders and instant visual press states.

## 3. Problem Prioritization Matrix

To scientifically select the most critical problems for the design proposal phase, each identified problem was evaluated on a scale from 1 to 5 using the following strict rubric:

### 3.1 Evaluation Rubric

- **Frequency:**

- **1:** < 10% users experienced it.
- **2:** 10% - 30% users experienced it.
- **3:** 30% - 50% users experienced it.
- **4:** 50% - 70% users experienced it (Very common).
- **5:** > 70% users experienced it (Almost everyone).

- **Severity (Based on Nielsen's Severity):**

- **1:** Cosmetic issue (Minor visual flaw, no real impact on task).
- **2:** Minor usability problem (Slight annoyance but easily overcome).
- **3:** Moderate problem (Causes hesitation or extra steps, but task is completed).
- **4:** Major problem (Severe disruption, user gets frustrated or task takes significantly longer).
- **5:** Usability catastrophe (Task blocked, user gives up or clicks away).

- **Evidence Strength:**

- **1:** Assumption or gut feeling (No data).
- **2:** Mentioned by only 1 interview participant, no survey data.
- **3:** Backed by minor survey data OR 2-3 interview clips.
- **4:** Backed by strong survey data AND clear interview observations.
- **5:** Overwhelming proof (High survey % AND repeatedly observed failing in video clips).

- **Feasibility:**

- **1:** Requires total system architecture rewrite (Very Hard).
- **2:** Requires major backend/API logic changes.
- **3:** Requires creating entirely new complex UI screens.
- **4:** Requires moderate frontend adjustments (e.g., adding sticky CSS, new filters).
- **5:** Quick fix (e.g., repositioning a button, changing text, fixing CSS limits - Very Easy).

$$\text{Priority Score} = \text{Frequency} + \text{Severity} + \text{Evidence Strength} + \text{Feasibility}$$

| Problem ID | Problem Description                                         | Frequency (1-5) | Severity (1-5) | Evidence Strength (1-5) | Feasibility (1-5) | Priority Score (4-20) | Decision        |
|------------|-------------------------------------------------------------|-----------------|----------------|-------------------------|-------------------|-----------------------|-----------------|
| P-01       | One-handed reachability issues with top-left hamburger menu | 4               | 5              | 5                       | 5                 | 19                    | Selected (High) |

| Problem ID | Problem Description                                                     | Frequency (1-5) | Severity (1-5) | Evidence Strength (1-5) | Feasibility (1-5) | Priority Score (4-20) | Decision          |
|------------|-------------------------------------------------------------------------|-----------------|----------------|-------------------------|-------------------|-----------------------|-------------------|
| P-02       | Lack of sticky search & filtering on long listing pages (Rating & News) | 5               | 4              | 5                       | 4                 | 18                    | Selected (High)   |
| P-03       | Overlapping video audio streams & lack of loading feedback              | 4               | 4              | 5                       | 5                 | 18                    | Selected (High)   |
| P-04       | Overloaded Hero section & lack of onboarding for first-time users       | 4               | 3              | 4                       | 5                 | 16                    | Selected (Medium) |

### 3.1 Scoring Rationale & Justification

#### 1. P-01 (One-handed navigation reachability - Score 19/20):

- *Frequency (4/5)*: 44.4% of surveyed users operate smartphones with one hand, and 43.1% reported physical effort/reachability issues.
- *Severity (5/5)*: Crucial navigation failure; top-left menu placement forces grip shifts and thumb stretching, creating high device drop risks during mobile commuting.
- *Evidence Strength (5/5)*: Validated by survey Q12/Q13 and direct video clip observations (P01, P02, P03).
- *Feasibility (5/5)*: Highly feasible to solve by introducing a persistent mobile Bottom Navigation Bar or floating action drawer.

#### 2. P-02 (Lack of sticky search & filtering on long lists - Score 18/20):

- *Frequency (5/5)*: 61.1% of users reported search difficulty when exact names are unknown; 81.9% requested search/filtering for long lists.
- *Severity (4/5)*: Forces 20+ seconds of repetitive vertical scrolling, severely degrading short 2-3 minute browsing sessions.
- *Evidence Strength (5/5)*: Validated by survey Q16/Q19/Q17 and interview clip observations (P01, P02, P04).

- *Feasibility (4/5)*: Feasible via sticky position CSS, query filter chips, and auto-suggest input handling.

### 3. P-03 (Video player feedback delay & audio overlap - Score 18/20):

- *Frequency (4/5)*: 52.8% of users expected old videos to auto-stop; 44.4% were confused by unresponsive video tap states.
- *Severity (4/5)*: Overlapping audio channels create extreme cognitive discomfort; missing loading spinners trigger repeated rage clicks.
- *Evidence Strength (5/5)*: Validated by survey Q20/Q21/Q22 and interview clip observations (P02, P03, P09).
- *Feasibility (5/5)*: Highly feasible via single-player event logic and skeleton loader components.

### 4. P-04 (Overloaded Hero section & onboarding gap - Score 16/20):

- *Frequency (4/5)*: 52.8% of surveyed participants were first-time users who had never heard of Freestyle Chess.
- *Severity (3/5)*: Disorients newcomers upon landing, though experienced users can scroll past it.
- *Evidence Strength (4/5)*: Validated by survey Q09/Q10 and interview clip observation (P03 L.T.K).
- *Feasibility (5/5)*: Highly feasible by reorganizing content hierarchy and adding a concise intro banner.

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## 4. Final Problem Statements

### Problem Statement 1 (Navigation & Reachability)

**Mobile users need** the primary navigation to sit within natural thumb reach **because** the hamburger menu is anchored top-left, forcing users to stretch their thumb or shift their grip just to tap a nav item, **especially when** browsing one-handed while on the move.

### Problem Statement 2 (Information Retrieval & Search)

**Mobile users need** a fixed, filterable way to search long content lists **because** the Rating list has no sticky search bar and News/events can't be searched by keyword, leaving users to scroll repetitively through long, unfiltered lists, **especially when** they don't remember the exact name or date they're looking for.

### Problem Statement 3 (Media Interaction & Feedback)

**Mobile users need** instant tap feedback and single-video playback **because** tapping a video gives no loading indicator and starting a new video doesn't pause the one already playing, causing overlapping audio, **especially when** watching match highlights or livestreams back-to-back.

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## 5. Conclusion & Transition to Project Proposal



The user analysis phase successfully consolidated the raw empirical data from Requirement 1 into four clear affinity clusters, prioritized the key problems using a structured score matrix, and formulated three final Problem Statements.

These problem statements will serve as the direct foundation for generating conceptual design solutions in **Requirement 3 (Project Proposal)**.